# Fraud Intelligence Pipeline & Analytics

End-to-End Data Engineering Case Study | Snowflake & Power BI

## 1. Executive Summary

This project showcases a high-performance, cost-optimized fraud detection system designed to handle enterprise-scale transaction data. By leveraging a **Medallion Architecture**, the pipeline transforms raw logs into actionable risk intelligence.

| 555,719 | $1,133,325 | $1.08 |
|---|---|---|
| Transactions Processed | Fraud Loss Identified | Total Build Cost |

## 2. The Risk Scoring Engine

Instead of relying on simple filters, I engineered a 4-dimensional scoring logic in the Snowflake Silver Layer:

- **Transaction Velocity:** Identifies "Rapid-Fire" fraud (multiple attempts in < 5 mins).
- **Amount Deviation:** Flags transactions significantly higher than a user's Z-score average.
- **Merchant Risk:** Weighted scores for merchants with historical high-frequency fraud.
- **Location Consistency:** Mismatch detection between customer profile and transaction geolocation.

## 3. Technical Implementation

| Technology | Role in Architecture |
|---|---|
| **Snowflake** | Data Warehousing, Streams, Tasks, and Stored Procedures (CDC & Scoring). |
| **Power BI** | 8-page interactive Command Center with Star Schema optimization. |

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|---|---|
| **SQL** / **DAX** | Window functions for velocity checks and measures for trend analysis. |

## 4. Key Insights

The analysis revealed that **Amount Deviation** is a 0.19 correlation predictor of fraud, outperforming simple time-based or location-based alerts. By implementing a 0.6 risk threshold, the system can automatically flag 30% of fraudulent attempts before they settle.